

Physical-time-adaptive nested low-rank transfer for phase-field evolution

1 MATERIAL TRAJECTORY

Target condition and phase-field evolution

PHASE-FIELD STATE



Target condition

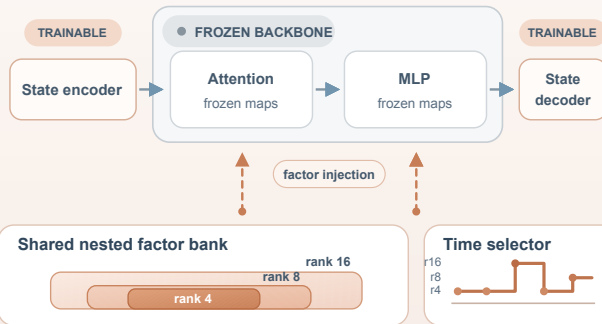
$\mu = (\varepsilon, \lambda, M, \text{anisotropy}, \dots)$

Phase-field dynamics

$E[q; \mu] \rightarrow \partial q / \partial \tau = N(q; \mu)$

2 ST-NLR TRANSFER ENGINE

Nested residuals are injected into a frozen multi-PDE backbone



3 MATERIAL-AWARE DEPLOYMENT

Choose the smallest material-feasible prefix

QUALITY CRITERIA



VALIDATION RULE

$$r_j^* = \min \mathcal{F}_j$$

ACTIVE PREFIX

r_j^*

PREDICTION

$\hat{q}_\tau$

one feasible prefix per physical time

TRAINING Joint multi-prefix calibration

rank-16 anchor

distill rank 8 / 4

material + spectral losses

DEPLOYMENT Frozen physical-time schedule

evaluate ranks 4 / 8 / 16

freeze rank schedule

activate one prefix